

**RAJALAKSHMI ENGINEERING COLLEGE**

**THANDALAM, CHENNAI-602105**

# CR23532 – CYBER AND DIGITAL FORENSICS

## CASE STUDY ASSIGNMENTS – II


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**COMPUTER SCIENCE AND ENGINEERING**

**(CYBER SECURITY)**

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# 2.1 SUSPECTED UNAUTHORIZED ACCESS TO A COLLEGE SERVER

**Scenario:** Out-of-hours copying of question-paper files from the CSE department examination server.

**Assumptions**

A1. The server runs Linux (e.g., Ubuntu) with SSH and a Samba file share. Authentication and file-access logs are enabled.

A2. The server clock is synchronised to an NTP source and set to IST (UTC+5:30). This must be verified, not assumed.

A3. The suspect workstation is a Windows 10 desktop in a staff room. It was found powered on.

A4. A USB flash drive was recovered from the workstation's front port.

A5. Written authorization to investigate has been obtained from the Principal/competent authority before any collection starts.

## 1. Forensic phases applied to this case

**Table 1. Investigation phases and activities**

| Phase | Activities in this case | Output |
|---|---|---|
| Identification | Log the administrator's complaint (who, what, when). Define scope (exam files, time window). Identify evidence sources: server, logs, workstation, USB, network devices, physical-access records. Confirm authorization. Rank sources by volatility. | Evidence source list, scope statement, authorization note |
| Collection | Document the scene with photographs. Capture volatile data from the running workstation (RAM, network connections, logged-in user) before shutdown. Image the disk and USB behind write blockers. Export server logs as read-only copies. Hash everything and start chain-of-custody forms. | Verified images, log copies, hash list, custody forms |
| Examination | Work only on forensic copies. Parse file system and log formats. Recover deleted files. Filter irrelevant files (known OS files via hash sets). Search for the exam file names and keywords. | Relevant artifacts extracted from raw data |
| Analysis | Correlate server logs, workstation artifacts and USB artifacts. Build the timeline. Test alternative explanations. Compare copied files with server originals using hashes. | Timeline, linkage of account to device to files, conclusions |
| Presentation | Prepare the report (technical and executive versions). Explain the method and limitations. Hand over evidence with custody records. Be ready to explain findings to the authority and, if needed, a court. | Final forensic report, presentation |

The phases are iterative. Examination may reveal a new source (e.g., a cloud-sync account), which sends the team back to Identification and Collection.

## 2. Forensic investigation flowchart

**COMPLAINT RECEIVED AND RECORDED**

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Confirm written authorization and define the investigation scope

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Identify evidence sources and rank them by volatility

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Secure and photograph the scene; isolate a live workstation from the network where appropriate

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DECISION: Is the workstation powered on?

YES → Capture volatile data (RAM, sessions, connections), then follow a controlled shutdown plan

NO → Keep powered off; document its state and do not boot it

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Acquire forensic images of the disk and USB through write blockers; preserve server logs

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Calculate and verify hashes; document chain of custody

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Examine verified copies; recover, filter, and search relevant artifacts

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Correlate artifacts, build a normalized timeline, and test alternative explanations

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Prepare report and plain-language summary; present findings and hand over evidence

**Figure 1. Investigation flow from complaint to presentation**

## 3. Potential digital evidence and investigative value

**Table 2. Evidence items and classification**

| ID | Evidence item | Source | What it can show | Investigative value | Volatility |
|---|---|---|---|---|---|
| E1 | Server authentication logs (e.g., SSH/system auth logs) | Server | Which account logged in, from which IP, at what time | High (primary): establishes the access event | Medium (rotation risk) |
| E2 | File-share/audit logs of the exam directory | Server | Which files were opened/copied/modified, by which account | High (primary) | Medium |
| E3 | Suspect workstation disk | Workstation | User profile, recent files, executed programs, USB connection traces, deleted files | High (primary) | Low (persistent) |
| E4 | Workstation RAM and network connections | Workstation (live) | Active sessions, running processes, remote connections | High but fragile | Very high |

| ID | Evidence item | Source | What it can show | Investigative value | Volatility |
|---|---|---|---|---|---|
| E5 | Recovered USB device | USB | Copied exam files, file-system timestamps, device serial | High (primary): possible exfiltration medium | Low |
| E6 | Copied exam files vs. server originals | USB / workstation | Hash match proves the copies came from the originals | Corroborative | Low |
| E7 | Network/firewall/DHCP logs | Network devices | Links an IP address to a device (MAC) at a given time | Corroborative | Medium |
| E8 | Physical access records (door log, CCTV, attendance) | Admin office | Presence of persons in the room at the time | Contextual / supporting | Medium (overwrite) |
| E9 | Administrator's complaint | Complainant | Starting point, reported time window | Contextual (lead) | Low |

Classification key:

- **Primary** evidence directly records the event.
- **Corroborative** evidence confirms a primary item from an independent source.
- **Contextual** evidence gives background or leads but cannot prove the act alone.

Volatility ordering follows RFC 3227: RAM and network state first, then logs and temporary files, then disk contents.

## 4. Precautions during collection and preservation

### Table 3. Precautions and their purpose

| Precaution | Why it matters |
|---|---|
| Obtain written authorization first | Keeps collection lawful and defensible |
| Photograph and sketch the scene, screen state, cables and USB ports | Proves the original state |
| Collect in order of volatility (RAM, network state, logs, disks) | Prevents loss of fragile data |
| Do not browse the suspect workstation or open files on it | Avoids changing timestamps and artifacts |
| Isolate from the network (disable the network port) rather than shutting down first if the system is live | Stops remote tampering while preserving RAM |
| Use hardware write blockers for the internal disk and USB | Prevents modification of the original media |
| Create a bit-for-bit forensic image; verify with hash (e.g., SHA-256) before and after | Demonstrates integrity |
| Export server logs by read-only copy, hash them, and record log-rotation settings | Prevents alteration and shows completeness |
| Record the clock offsets (server, workstation, network devices) versus a trusted reference | Needed for a reliable timeline |
| Label, bag (anti-static for drives), seal and store evidence securely | Prevents tampering and damage |
| Maintain a continuous chain-of-custody log | Shows who held the evidence at each stage |
| Perform all analysis on verified copies; keep originals sealed | Original is the master record |
| Maintain contemporaneous notes of every action with date/time and examiner name | Allows the work to be reproduced and defended |

## 5. Establishing the timeline of unauthorized access

### Method

1. List every timestamp source: server auth log, share audit log, workstation file-system timestamps (created/modified/accessed), registry/event artifacts (USB first/last connected), USB file timestamps, network logs, door/CCTV logs.
2. Record each source's clock offset and time zone, and convert all entries to a single reference (e.g., UTC, with IST shown alongside).
3. Merge entries into one sorted table with columns: time, source, event, account/device, confidence.
4. Identify the *access window* from the server logs (login, file access, logout).
5. Look for the matching events on the workstation (remote session, file copy, USB insertion) inside that window.
6. Corroborate with independent sources (network logs, physical access records).
7. Mark gaps and conflicts rather than hiding them.

**Table 4. Sample timeline structure**

| Time (IST) | Source | Event | Account/Device | Role in analysis |
|---|---|---|---|---|
| 21:58 | Door/attendance log | Room entry recorded | Card ID *X* | Contextual |
| 22:03 | Network/DHCP log | IP *a.b.c.d* leased to workstation MAC | Workstation | Links IP to device |
| 22:07 | Server auth log | Successful login from *a.b.c.d* | User *U* | Primary |
| 22:10–22:14 | Share audit log | Exam directory files opened/read | User *U* | Primary |
| 22:12 | Workstation registry/event artifact | USB device connected (serial *S*) | Workstation | Corroborative |
| 22:15 | USB file-system data | Files with exam names created | USB | Corroborative |
| 22:19 | Server auth log | Session logout | User *U* | Primary |

Timestamps from different systems need alignment. A small clock offset can change the order of events.

## 6. Final forensic report and presentation to a non-technical authority

### Sample forensic report structure

1. Cover page: case ID, title, examiner, date, classification (confidential)
2. Executive summary (half page, plain language)
3. Authorization and scope
4. Background and complaint
5. Evidence inventory (item, ID, description, hash, source)
6. Methodology and tools (versions, write blockers, hashing algorithm)
7. Chain-of-custody summary
8. Findings (each finding: what, where, evidence ID, confidence)

9. Timeline of events
10. Analysis and alternative explanations considered
11. Limitations and assumptions
12. Conclusion (facts supported by evidence)
13. Recommendations (e.g., log retention, access control, USB policy)
14. Appendices: hash lists, custody forms, tool outputs, glossary, examiner declaration

**Presenting to a non-technical authority**

- Open with a one-paragraph answer: what happened, how sure we are, and what is not known.
- Use plain language and everyday analogies (e.g., a login record is like a visitor register).
- Show one visual timeline rather than raw logs.
- Separate **facts** (supported by evidence), **interpretation**, and **opinion**.
- State confidence levels and limitations honestly, avoiding statements like "the user is guilty". The report states what the evidence shows.
- Offer recommendations the authority can act on, such as policy changes.
- Keep technical detail in appendices for those who want it.

## 7. Legal and ethical considerations

**Table 5. Legal and ethical considerations (college environment)**

| Area | Consideration |
|---|---|
| Authority | Act only with written authorization from a competent authority; institutional IT/acceptable-use policy should permit inspection of institutional systems |
| Legal framework (India) | Unauthorized access and copying may fall under the Information Technology Act, 2000 (e.g., Sections 43 and 66). Admissibility of electronic records is governed by Section 63 of the Bharatiya Sakshya Adhiniyam, 2023, which replaced the Indian Evidence Act, 1872 from 1 July 2024. The Section 63 certificate must disclose the hash value of the electronic record, so hashing at collection is a legal requirement in practice, not only good practice |
| Privacy and data protection | The workstation and server may hold students' personal data. Collect and examine only what is relevant, and follow the Digital Personal Data Protection Act, 2023 |
| Proportionality | Do not examine personal files unrelated to the complaint |
| Confidentiality | Share findings on a need-to-know basis; protect exam material and personal data |
| Impartiality | Examiner must remain neutral, record exculpatory as well as incriminating evidence, and avoid presuming guilt |
| Chain of custody and integrity | Required for evidence to be accepted and trusted |
| Escalation | If criminal conduct is likely, involve law enforcement and legal counsel early |

## Conclusion

A defensible investigation of the server incident depends on authorized, ordered collection, verified imaging, correlation of server and workstation evidence on a normalised timeline, and a report that separates facts from interpretation. Hashing and the chain of custody make the evidence reliable and defensible. Any attribution to a person must acknowledge limits such as shared credentials and clock error.

## References


1. Kent, K., Chevalier, S., Grance, T., Dang, H. (2006). *Guide to Integrating Forensic Techniques into Incident Response*, NIST SP 800-86.
2. Brezinski, D., Killalea, T. (2002). *Guidelines for Evidence Collection and Archiving*, RFC 3227, IETF.
3. ISO/IEC 27037:2012. *Guidelines for identification, collection, acquisition and preservation of digital evidence*.
4. Casey, E. (2011). *Digital Evidence and Computer Crime*, 3rd ed., Academic Press.
5. Information Technology Act, 2000 (India); Bharatiya Sakshya Adhiniyam, 2023, Section 63; Digital Personal Data Protection Act, 2023.

# 2.2 LINKING A SUSPICIOUS FILE TO A USER USING FILE-SYSTEM EVIDENCE

**Selected scenario:** Attributing a disguised executable and an encrypted archive to one of several staff accounts on a shared lab desktop.

**Assumptions**

A1. The desktop runs Windows 10 on an NTFS-formatted internal drive with three standard local accounts (**staffA, staffB, staffC**) and a local administrator account. All names are **hypothetical**.

A2. A forensic image has already been acquired with write protection and verified by hash.

A3. The suspicious items are an executable with a document-style name, several files with the hidden attribute, an encrypted archive, and browser profile data.

A4. ILookIX is the analysis tool. Menu names vary by version, so the workflow below describes functions, not exact screens.

## 1. Hardware and software components to document before analysis

**Table 1. System inventory table**

| Category | Component to document | Why it matters | Example entry (Hypothetical placeholder) |
|---|---|---|---|
| Hardware | Make, model, serial number, asset tag | Identifies the exact device | *Brand/Model, Serial 1234* |
| Hardware | CPU, RAM size | Context for memory capture and performance | *Specs* |
| Hardware | Storage devices: type (HDD/SSD), capacity, interface, model, serial | Identifies source media; SSD behaviour (TRIM) affects deleted-data recovery | *1 x SSD, 256 GB* |
| Hardware | Partition scheme (MBR/GPT), partitions, unallocated/hidden areas | Hidden partitions can hold evidence | *GPT, 3 partitions* |
| Hardware | External ports and attached/previously attached devices | USB and external storage leads | *4 x USB* |
| Hardware | Network adapter(s), MAC addresses | Links to network logs | *MAC (recorded)* |
| Hardware | BIOS/UEFI date/time versus reference time | Clock accuracy for timestamps | *Offset recorded* |
| Software | OS name, version, build, install date | Determines artifact locations and formats | *Windows 10, build recorded* |
| Software | Time zone and regional settings | Needed to interpret timestamps | *IST* |
| Software | File system(s) | NTFS provides $MFT, timestamps and ADS | *NTFS* |
| Software | User accounts (names, SIDs, last logon) | Core of attribution | *staffA, staffB, staffC* |
| Software | Installed applications, browsers, encryption/archiving tools | Explains the executable and encrypted data | *List from system* |
| Software | Security settings: BitLocker/EFS, antivirus, audit policy | Affects accessibility and logging | *Status recorded* |

| Category | Component to document | Why it matters | Example entry (Hypothetical placeholder) |
|---|---|---|---|
| Forensic tooling | Write blocker model, imaging tool and ILookIX versions, hashing algorithm | Reproducibility and defensibility | *Versions recorded* |

## 2. How storage devices, operating systems, file systems and metadata help locate evidence

### Table 2. Layers that guide the search

| Layer | What it contains | How it helps locate evidence |
|---|---|---|
| Storage device | Physical sectors, partitions, unallocated space | Shows what exists beyond visible files: hidden partitions, deleted data. SSD TRIM may limit recovery |
| Operating system | Per-user profile folders, registry hives (e.g., NTUSER.DAT), event logs, Prefetch, recent-file and shortcut (LNK) data, Jump Lists | Records user and system activity, giving the "who used what" context |
| File system (NTFS) | Master File Table ($MFT) with timestamps, file owner/permissions, alternate data streams, journals ($UsnJrnl, $LogFile) | Shows file existence, paths, creation/modification/access/MFT-change times, and often deleted-file remnants |
| File metadata | Document author/last-saved-by, application-specific data, executable header details, image EXIF | Links content to a creator, tool, time or device |
| Application data | Browser history, downloads, cache, cookies | Shows where a file came from and how it was used |

**Practical use:** start with the user-profile paths and timestamps to narrow the search, then use metadata and registry artifacts to confirm who interacted with a file.

## 3. Hidden and encrypted files: effect on the investigation and handling

### Table 3. Hidden and encrypted content

| Item | Effect on investigation | Appropriate handling |
|---|---|---|
| Files with hidden/system attribute | Not shown in normal views, so easily missed. The attribute is only a flag, not proof of concealment intent | Include hidden items in all listings. Record the attribute without drawing intent conclusions |
| Renamed extensions (e.g., executable named like a document) | Hide true type from casual inspection | Run file-signature (header) analysis to detect mismatches |
| Alternate data streams and slack/unallocated space | Can hold data not visible in the directory listing | Examine through file-system-aware tools. Carve if needed |
| Encrypted containers/archives/documents | Content inaccessible. Keyword search and hash-set filtering fail. Metadata (name, size, timestamps, path) is still available | Preserve and hash the encrypted file. Do not alter it. Document its properties. Look for lawful ways to obtain keys (e.g., stored credentials, key material in memory capture, or legal disclosure) |
| Whole-volume or user-level encryption (e.g., BitLocker, EFS) | May make the image unreadable without keys | Check for encryption before powering off a live system. Capture RAM where possible |
| Password cracking or compelled disclosure | May be time-consuming and raises legal issues | Do so only under proper authorization. Take legal advice on compelled disclosure |

**General handling points:** record everything, preserve originals, work on copies, do not conclude intent from a hidden or encrypted state alone, and separate "present" from "used by whom".

## 4. Evidence-linkage matrix

**Table 4. Evidence-linkage matrix: correlating artifacts to a user (Hypothetical examples; actual findings depend on the real image)**

| # | Artifact | Where found | What it records | Link to user | Strength |
|---|---|---|---|---|---|
| L1 | File path of the suspicious executable | File-system listing | Location, e.g., inside a specific user's profile folder (*C:\Users\staffB\...*) | Placement in a profile folder suggests that account | Moderate |
| L2 | NTFS owner / security descriptor | File properties / $MFT | Owner SID of the file | Maps to an account via SID | Moderate: shows the account that created it, not the person |
| L3 | Created/modified/accessed timestamps | $MFT | When the file appeared and changed | Compared with that account's logon sessions | Supporting |
| L4 | Registry execution history (e.g., UserAssist, RunMRU) in that user's NTUSER.DAT | User registry hive | Programs run by that account, run count, last run | Per-account evidence of execution | Strong |
| L5 | Prefetch entry for the executable | Windows Prefetch | Executable ran on the system, and when | Machine-wide: does not name the user | Supporting |
| L6 | Recent items/LNK/Jump List for the archive or files | User profile | Files opened by that account | Per-account interaction | Strong |
| L7 | Security event log logon/logoff events | Windows event logs | Which account was logged on and when | Places an account at the console in the time window | Strong when timed with L3–L6 |
| L8 | Browser history/downloads/cache | That user's browser profile | Sites visited, files downloaded (source URL, time, save path) | Origin of the file, tied to the profile | Strong for download evidence |
| L9 | Document metadata (author, last saved by) | Inside document files | Name or username embedded by the application | Can name a creator | Moderate. Names can be edited or copied |
| L10 | Encrypted archive properties | File system | Name, size, timestamps, location | Shows existence and location; content unknown | Supporting |

**Correlation principle:** one artifact rarely proves attribution. Confidence grows when independent artifacts agree on account, time and path.

## 5. Workflow for importing/analyzing evidence in ILookIX

**Figure 1. ILookIX analysis workflow**

CASE SETUP: Record case ID, examiner, authorization, and scope

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EVIDENCE IMPORT: Acquire using an approved workflow or import a verified forensic image/ASB container

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INTEGRITY CHECK: Compare the image hash with the acquisition record

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PROCESSING: Parse file systems, list active/deleted files, hash files, apply known-file filtering, check signatures, index keywords, and extract relevant artifacts

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TRIAGE AND REVIEW: Filter by user profile, hidden/system attributes, signature mismatches, and encrypted-file indicators

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ARTIFACT ANALYSIS: Review registry artifacts, event logs, LNK/Jump Lists, browser data, and other relevant records

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CORRELATION: Compare timestamps and account activity; normalize time zones and document uncertainty

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FINDINGS AND REPORT: Bookmark findings with notes; report hashes, tool/version details, and limitations

signature mismatch, encrypted-file indicators

**Step notes**

- **Step 1:** the source media is never touched. All analysis runs on the image.

- **Step 2:** the matching hash demonstrates the image was not altered after acquisition.
- **Step 3:** hash-set filtering reduces noise from standard OS files. Signature analysis exposes the executable disguised as a document.
- **Step 4:** encrypted files are identified by signature or properties and are preserved without modification.
- **Step 5:** each bookmark records what it is, where it was found and why it matters, supporting reproducibility.
- Note the ILookIX version and the exact functions used in the report. Do not claim results that were not generated.

## 6. Direct linking artifact vs. supporting/contextual artifact

**Table 5. Direct vs. supporting artifacts**

| Aspect | Directly linking artifact | Supporting/contextual artifact |
|---|---|---|
| Definition | Ties a specific account to a specific action (create, run, download, open) | Shows that something existed or was possible, without tying an account to an action |
| Examples (this case) | Execution record for the executable in *staffB*'s own registry hive. Browser download record in *staffB*'s profile for the file. Recent-items entry in *staffB*'s profile for the archive | File simply residing in a shared folder. File timestamps alone. Prefetch entry (machine-wide). Hidden attribute. Installed encryption software. Machine logon log without activity detail |
| Strength | Strong, but still account-level | Weak alone, useful for context and corroboration |
| Use in report | Primary finding | Background or corroboration |

Even a direct artifact links an **account** to an activity, not necessarily a **person**. Linking the account to a person needs additional evidence such as physical presence or authentication records.

## 7. Limitations and alternative explanations

### Table 6. Limitations to consider before attribution

| # | Limitation / alternative explanation | Why it matters | How to address it |
|---|---|---|---|
| 1 | Shared or weak credentials; the account owner may not be the actual user | An account is not a person | Seek corroboration: access logs, physical presence, other devices |
| 2 | Malware or remote access could have created or run the files | Activity could occur without the user's knowledge | Examine for malicious or remote-access software, and external connections |
| 3 | Timestamps can be changed (e.g., timestamp manipulation), copied, or altered by the clock/time zone | Weakens timeline conclusions | Compare the $STANDARD_INFORMATION and $FILE_NAME times, journals and event logs; document clock offsets |
| 4 | Files can be copied from another machine or USB, preserving original times | Creation time may not mean creation on this device | Check for USB artifacts and download records |
| 5 | Administrator or another user could have placed files in someone else's profile | Path alone does not prove ownership | Use execution and logon artifacts together with path |
| 6 | Encrypted content cannot be read | Purpose and content remain unknown | State this in the report. Avoid assumptions about content |
| 7 | Deleted data may be overwritten and tool interpretation can err | Incomplete picture | Validate with a second tool or method; record tool versions |

## Conclusion

The suspicious executable and encrypted archive can be linked to a user account by correlating file paths, NTFS owner information, timestamps, registry execution traces, event-log logons and browser records. The strongest case comes from several independent, per-account artifacts that agree in time. Limitations such as shared credentials, malware, copied timestamps and unreadable encrypted content must be stated, so the report attributes the activity to an *account* with a stated level of confidence, not to a named *person* without further evidence.

### References


1. Carrier, B. (2005). *File System Forensic Analysis*. Addison-Wesley.
2. Casey, E. (2011). *Digital Evidence and Computer Crime*, 3rd ed., Academic Press.
3. Kent, K. et al. (2006). *Guide to Integrating Forensic Techniques into Incident Response*, NIST SP 800-86.
4. ISO/IEC 27037:2012. *Guidelines for identification, collection, acquisition and preservation of digital evidence*.
5. ILookIX / IXImager documentation (Perlustro LP): consult the version installed in your lab and cite its exact version.

# 2.3 FORENSIC IMAGING AND SAFEKEEPING OF A SUSPECT STORAGE DEVICE

**Selected scenario:** Imaging a seized laptop and an attached USB drive in a suspected research-data theft.

**Assumptions**

A1. A laptop with one internal SATA SSD/HDD and one attached USB flash drive is seized from a staff desk. Both are covered by a written seizure authorization.

A2. The laptop was found powered off, lid closed, with the USB drive plugged in. If it were on, RAM capture would come first (see Table 1, step 5).

A3. Hardware write blockers, a sanitised destination drive, and IXImager (ILookIX suite) are available in the lab.

A4. IXImager menu names vary by version, so the workflow describes functions, not exact screens. The ASB container is described by purpose; confirm its exact fields in your installed version.

A5. All identifiers (serials, case numbers, names, dates) in tables are **hypothetical placeholders**.

## 1. Procedure to collect and transport the laptop and USB device

**Table 1. Collection and transport procedure**

| Step | Action | Purpose |
|---|---|---|
| 1 | Confirm authorization; open a case number; assign an evidence collector | Lawful, documented start |
| 2 | Photograph the scene and the devices in place (screen, ports, cables, USB position, labels) | Records original state |
| 3 | Note state: powered on/off, lid, power source, visible connections | Decides live collection vs. shutdown handling |
| 4 | Do not power on, log in, or open files | Avoids changing timestamps, logs and artifacts |
| 5 | If found powered on: capture RAM and note running state, check for disk encryption (e.g., BitLocker), then follow a controlled shutdown plan | Volatile data and encryption keys disappear at power-off |
| 6 | Remove the USB drive only after photographing it in place; do not open it | Preserves the original attachment state |
| 7 | Record make, model, serial number and asset tag of the laptop, internal drive (if accessible) and USB drive | Unique identification |
| 8 | Disconnect power as needed and prevent wireless connections (e.g., Faraday bag); only trained staff should remove internal network cards | Prevents remote wiping or modification |
| 9 | Place each item in a separate anti-static bag; seal with tamper-evident tape; label with case number, item ID, date/time, collector's name and signature | Prevents mix-ups and shows tampering |
| 10 | Pack with padding in a rigid container; keep away from magnets, heat and moisture | Prevents physical damage and data loss |
| 11 | Start the chain-of-custody form at the seizure point; every hand-over is signed | Continuous accountability |
| 12 | Transport directly to the lab; do not leave the items unattended; log departure and arrival times | Prevents unauthorized access |

| Step | Action | Purpose |
|---|---|---|
| 13 | On arrival: check seals against the form, photograph, log condition, store in a locked evidence room | Confirms integrity at receipt |

## 2. Physical acquisition and why analysis uses an image

**Physical acquisition** is a sector-by-sector (bit-stream) copy of the entire storage device, including the partition table, unallocated space, file-system slack space, deleted-file remnants and hidden areas. A logical copy captures only the files the operating system presents, so it misses much of this.

**Why analyse the image and not the original:**

- **Integrity:** the original stays untouched. Even booting the system or mounting the drive normally changes timestamps and logs.
- **Repeatability:** any examiner can re-run the analysis on a verified copy and get the same results.
- **Risk:** a drive may fail, so analysing the original risks irreversible loss. A verified image can be copied again.
- **Defensibility:** courts and reviewers expect the original to remain unaltered, with the image proven identical by hash.
- **Safety:** analysis tools may crash or write temporary data. This does no harm to a copy.

**Caveats:** physical acquisition of SSDs may not capture data already erased by TRIM or garbage collection. If the drive is encrypted, the image is as unreadable as the original unless keys are available.

## 3. Imaging workflow using IXImager

**Figure 1. Acquisition flowchart**

AUTHORIZATION: Open the case and confirm authority to seize and examine the devices

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IDENTIFY SOURCE: Record device make, model, serial number, capacity, and condition

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PREPARE DESTINATION: Use sanitized destination media with sufficient capacity

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PROTECT SOURCE: Connect through a tested hardware write blocker and confirm read-only status

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CONFIGURE IXIMAGER: Record case details, tool version, time references, output format, and hash settings

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ACQUIRE IMAGE: Image the physical device and record progress, read errors, and bad sectors

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VERIFY INTEGRITY: Compare source and image hashes; investigate any mismatch and repeat acquisition if appropriate

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PRESERVE: Complete logs, create a verified working copy, seal originals, and update chain of custody

**Table 2. Imaging procedure (IXImager)**

| Stage | Detailed steps | Record in the acquisition log |
|---|---|---|
| Source identification | Read the device label and compare with the seizure record. Connect through the write blocker and note how the device is reported (model, serial, capacity, sector count) | Source description, serial, capacity |
| Write protection | Use a hardware write blocker (preferred). Check indicators and confirm the source shows as read-only before imaging. For USB, use a USB write blocker, not only an operating-system setting | Blocker make/model/firmware, test result |
| Destination preparation | Use a forensically sanitised drive with enough free space. Format it and record the file system | Destination serial, sanitisation method |
| Tool setup | Start IXImager. Select the physical device as the source (not a logical volume). Enter case number, evidence number, examiner and description. Choose the ASB container format and options such as compression and verification, where offered | Tool version, settings |
| Imaging | Start acquisition. Monitor progress. Note any read errors or bad sectors reported and how the tool handles them | Start/end time, errors, sectors affected |
| Verification | Let the tool verify the image or recompute its hash. Compare against the hash of the source calculated during acquisition | Hash algorithm, source hash, image hash, match result |
| Hashing | Use a strong algorithm (SHA-256, optionally with MD5 for compatibility). Record in the log and on the custody form | Hash values (record real values only from the tool) |
| Documentation | Complete the acquisition log, photograph the setup, sign and date. Make a working copy for analysis, and keep the master sealed | Examiner, date/time, time zone, notes |

Repeat the same procedure for the USB drive, as a separate evidence item with its own ID.

## 4. Hash values and integrity before and after acquisition

A hash function converts data of any size into a fixed-length value. Any change to the input, even one bit, gives a completely different output, so a hash acts as a digital fingerprint of the data.

**Table 3. Use of hash values through the evidence lifecycle**

| Point | Action | What a match shows |
|---|---|---|
| Before/during acquisition | Calculate the hash of the source device (through the write blocker) | Baseline fingerprint of the original |
| Immediately after acquisition | Hash the image and compare with the source hash | Image is a bit-exact copy |
| On transfer or copy | Re-hash the image after copying to a new drive or sending it | Copy was not corrupted or altered in transit |
| Before analysis | Verify the image hash again at the start of examination | Evidence is unchanged since acquisition |
| Before reporting/court | Verify again and quote the hash in the report | Integrity is demonstrated at every stage |

- A mismatch means something changed or failed (corruption, bad sectors, or tampering). It must be investigated and explained, not ignored.

- If the source has unreadable sectors, repeated reads may not give identical hashes. Record this in the log, since it explains why the match may not be perfect.
- MD5 and SHA-1 have known collision weaknesses, so SHA-256 is preferred. Recording two algorithms gives extra assurance.
- In India, the certificate under Section 63 of the Bharatiya Sakshya Adhiniyam, 2023 requires the hash value of the electronic record, so hashes must be recorded at acquisition (see Assignment 2(1)).

## 5. Role of the ASB container

An **ASB container** is the evidence-image format used by the ILookIX/IXImager suite. It packages the acquired data together with descriptive and integrity information, so that evidence can be stored, moved, verified and imported into ILookIX as a single managed item. Confirm exact capabilities (compression, segmentation, metadata fields, any protection options) in the documentation for your installed version.

**Table 4. Information that should accompany the evidence**

| Item | Detail |
|---|---|
| Case and evidence IDs | Case number, evidence/item number |
| Source device description | Make, model, serial, capacity, interface, where seized |
| Examiner and authority | Examiner name, organisation, authorization reference |
| Acquisition details | Tool name and version, date/time and time zone, write blocker used, settings (format, compression) |
| Integrity data | Hash algorithm(s) and values for source and image; verification result |
| Error log | Bad sectors/read errors and how they were handled |
| Time references | Device clock vs. reference time at acquisition |
| Chain-of-custody record | Copy of the form and storage location |
| Copies | Number, location, and hash of each copy |

## 6. Data carving

**Data carving** recovers files from raw data by searching for known file signatures (headers, and where defined, footers) instead of relying on file-system records. Examples: a JPEG begins with the bytes FF D8 FF and ends with FF D9. A PDF begins with "%PDF" and ends with "%%EOF".

### When carving is needed

- The file system is damaged, corrupted or reformatted, so directory entries and the master file table are gone.
- Files were deleted and their metadata overwritten, but content remains in unallocated space.
- The partition table is missing or corrupted.
- Data sits in unallocated space, slack space or a memory/page file.

### Limitations

- Carved files usually lose file names, paths and timestamps.

- Fragmented files may be recovered only partly or wrongly joined.
- Carving can produce false positives and unrelated files, so results must be validated (open the file, check the hash, compare context).
- Files without clear signatures are hard to carve.

## 7. Date and time problems in digital evidence

### Table 5. Time issues and how to document them

| Problem | Why it matters | How to document |
|---|---|---|
| Time zone differences (local time vs. UTC) | Systems store times differently; NTFS keeps UTC, FAT keeps local time | Record each source's time zone and convert to one reference |
| Incorrect system clock or BIOS time | Timestamps may be shifted | At acquisition, record the device clock beside a trusted reference time (e.g., a checked clock or NTP source) and the offset |
| Daylight saving changes | One-hour shifts | Note whether DST applied and when |
| Different timestamp resolution (e.g., FAT 2-second resolution) | Small differences may be artifacts | Note file-system characteristics |
| Timestamps changed by copy, move, or extraction | Creation time may reflect the copy, not the origin | Interpret together with other artifacts; avoid single-timestamp conclusions |
| Last-access times not always updated | Absence of an update does not mean no access | State this limitation |
| Deliberate manipulation | Times can be altered by the user | Compare multiple sources (journals, event logs, logs on other devices) |

**Documentation practice:** write the device time and reference time with the offset in the acquisition log, state the time zone used in every table of the report, and describe uncertainty in plain terms (e.g., "times shown are UTC+5:30 after correcting a clock offset of N minutes noted at acquisition").

## 8. Chain-of-custody form and table for the seized laptop

### Table 6. Chain-of-custody form (header)

| Field | Entry (Hypothetical placeholder) |
|---|---|
| Case number | CASE-1056 |
| Evidence item ID | LT-01 (laptop); HD-01 (internal drive); USB-01 (USB drive) |
| Description | Laptop, make/model, serial XXXX, colour, condition, seal number |
| Seized from / location | Staff desk, room/department |
| Seized by | Collector name, designation |
| Authorization reference | Written authorization no. and date |

### Table 7. Chain-of-custody table (Hypothetical sample entries; replace with your own)

| # | Date/time | Item | Released by | Received by | Purpose / location | Condition / seal | Signatures |
|---|---|---|---|---|---|---|---|
| 1 | D1 10:15 | LT-01, USB-01 | Room staff / scene | Collector (seizure) | Seizure at scene | Bagged and sealed, seal no. S1 | Both |
| 2 | D1 11:00 | LT-01, USB-01 | Collector | Transport officer | Transport to lab | Seals intact | Both |

| # | Date/time | Item | Released by | Received by | Purpose / location | Condition / seal | Signatures |
|---|---|---|---|---|---|---|---|
| 3 | D1 12:10 | LT-01, USB-01 | Transport officer | Evidence custodian | Receipt at lab, locked store | Seals checked, photographed | Both |
| 4 | D2 09:30 | LT-01 (internal drive removed), USB-01 | Evidence custodian | Forensic examiner | Imaging | Seals opened, new seal no. noted | Both |
| 5 | D2 15:45 | HD-01, USB-01, images | Forensic examiner | Evidence custodian | Return to storage; images stored | Originals resealed, new seal numbers | Both |
| 6 | D3 10:00 | Working copy of image | Evidence custodian | Examiner | Laboratory examination | Hash verified before use | Both |

Each time a seal is broken or re-applied, record the seal number, reason and who was present.

## References


1. Kent, K. et al. (2006). *Guide to Integrating Forensic Techniques into Incident Response*, NIST SP 800-86.
2. ISO/IEC 27037:2012. *Guidelines for identification, collection, acquisition and preservation of digital evidence*.
3. Brezinski, D., Killalea, T. (2002). *Guidelines for Evidence Collection and Archiving*, RFC 3227.
4. NIST (2015). *Secure Hash Standard (SHS)*, FIPS PUB 180-4.
5. Carrier, B. (2005). *File System Forensic Analysis*. Addison-Wesley.
6. Casey, E. (2011). *Digital Evidence and Computer Crime*, 3rd ed., Academic Press.
7. ILookIX / IXImager documentation (Perlustro LP): cite the version in your lab.

# 2.4 RECOVERY OF DELETED EVIDENCE FROM VOLUME SHADOW COPIES

**Selected scenario:** Recovering deleted internship-report evidence and browser traces from Volume Shadow Copies.

**Assumptions**

A1. A verified forensic image of a Windows 10 NTFS system is available. The suspect is an employee, referred to as **user E** (hypothetical).

A2. System Protection (restore points) was enabled, so Volume Shadow Copies (VSCs) exist on the system volume.

A3. The target is a document (*Internship_Report_Final.docx*, hypothetical name) and browser history for a specific site.

A4. All dates, times and file names are **hypothetical**. No real recovery results are claimed.

## 1. Selecting and categorizing evidence before detailed analysis

An image can hold hundreds of thousands of files. Investigators narrow it to what is relevant without destroying anything, since the full image is preserved.

**Selection procedure**

1. **Re-read the scope** from the authorization: user, device, time window, type of activity.
2. **Verify the image hash** and record tool versions.
3. **Exclude known files:** apply hash sets of standard operating-system and application files to filter noise.
4. **Filter by user, path and date range** (e.g., user E's profile, the period around the suspected activity).
5. **Filter by type and content:** documents, archives, browser data, keyword and file-name searches.
6. **Check for disguised files** using file-signature analysis.
7. **Include deleted and recovered items** (Recycle Bin, unallocated space, shadow copies).
8. **Tag and categorize** each item, record why it was included or excluded, and keep the filtering criteria in the notes.

Excluded data is *filtered from view*, not deleted. The reasoning is recorded so a reviewer can check it.

## 2. Evidence classification scheme

**Table 1. Evidence classification table**

| Category | Definition | Examples in this case | Evidential weight / handling |
|---|---|---|---|
| User-created files | Content created or edited by the user | Report documents, spreadsheets, notes, saved images | High: examine content and metadata; compare versions |
| System/application files | Files generated by the OS or applications | Windows event logs, registry hives, Prefetch, application settings, Recycle Bin index files | Medium to high: show activity context (what ran, who was logged in, when) |
| Potentially suspicious files | Items whose type, location or behaviour is unusual | Renamed or mismatched extensions, encrypted archives, wiping/cleaning utilities, unknown executables | Needs investigation: do not assume intent; validate |
| Deleted/recovered artifacts | Items no longer visible in the normal file system but recovered | Files in Recycle Bin, carved files, older versions in shadow copies, deleted browser entries | High if validated; always record source (snapshot ID, offset) |
| Supporting/contextual evidence | Background that explains or corroborates | Installed software list, user accounts, USB device history, logon events, network/cloud sync settings | Supporting: strengthens other findings but rarely proves them alone |
| *Excluded/irrelevant* | Known system files or clearly out-of-scope data | Standard Windows files matched by hash set | Not analysed further; exclusion logged |

## 3. Event Analysis for reconstructing activity

**Event Analysis** brings time-stamped records from many artifacts into one combined, sorted view so the sequence of actions becomes visible. The exact feature set depends on your ILookIX version; the general method is:

1. Collect events: file creation/modification/deletion times, Recycle Bin entries, registry key times, event-log entries, browser visits and downloads, USB connection times, program execution records.
2. Normalise all times to one reference time zone.
3. Sort by time and filter around the period of interest.
4. Group related events into sequences (e.g., browse, download, create file, copy to USB, delete).
5. Look for clusters, gaps (possible log clearing), and contradictions.
6. Validate key events using at least two independent artifacts.

Example use here: Event Analysis can show that a document was last modified, then a USB device connected, then the document deleted, then the browser history cleared, all within minutes. This sequence suggests an *activity pattern* but does not, by itself, prove who did it or why.

## 4. Cloud Analysis and Lead Analysis

**Table 2. Purpose and questions answered**

| Technique | Purpose | Questions it may help answer |
|---|---|---|
| Cloud Analysis | Identify artifacts showing use of cloud storage, email or online services on the system, such as sync-client folders, account identifiers, cloud URLs in browser history, and cloud-related application files | Was the document uploaded or synced to a cloud account? Which cloud services and accounts did the user use? When did syncing occur? Is more evidence held outside the device and needs a legal request or separate authorization? |
| Lead Analysis | Extract and rank pieces of information (e.g., names, email addresses, URLs, phone numbers, keywords, user names) that point to further lines of enquiry and relationships between items | Who did the user communicate with? Which accounts, sites or contacts appear repeatedly? Which files or keywords connect to the incident? What should the investigator examine next? |

Cloud Analysis helps find *where else* evidence might exist. Lead Analysis helps decide *what to examine next*. Both give leads that must be confirmed with direct evidence. Accessing cloud accounts requires proper legal authority.

## 5. How Volume Shadow Copy analysis helps

**Volume Shadow Copy Service (VSS)** is a Windows feature that creates point-in-time snapshots of an NTFS volume, using block-level storage of changes in the System Volume Information area. Snapshots are created, for example, by restore points, software installation or scheduled backups.

**Value to the investigation**

- **Recovery:** a file deleted or overwritten after a snapshot may still exist in its older state inside that snapshot.
- **Comparison:** the same file or registry/browser database can be compared across snapshots and the current volume to show what changed (e.g., content edited, history entries removed).
- **Time anchoring:** each snapshot has a creation time, showing the system's state at that moment.
- **Anti-forensic detection:** the absence of expected snapshots, or signs that snapshots were deleted, is itself a finding.

**Limits:** snapshots exist only if they were created and not deleted or removed automatically when storage limits were reached. Snapshots reflect state at one moment, so activity between snapshots may be missed.

**Figure 1. Volume Shadow Copy recovery workflow**

VERIFY: Verify the forensic image hash and record tool versions

↓

DETECT: Determine whether Volume Shadow Copies are present on the relevant volume

↓

INVENTORY: Record each snapshot identifier, creation time, and available metadata

↓

EXAMINE READ-ONLY: Open snapshots using a supported forensic tool or documented workflow

↓

COMPARE: Compare snapshots with the current volume and each other; identify missing/changed files and earlier browser data, registry hives, or logs

↓

EXTRACT AND HASH: Extract relevant items with metadata and calculate hashes for exported artifacts

↓

VALIDATE: Check provenance, metadata, content, consistency, and alternative explanations

↓

CORRELATE AND REPORT: Record snapshot ID and path; correlate with the timeline and document limitations

snapshot)

## 6. Hypothetical timeline: create, modify, copy, delete, recover

**Table 3. Hypothetical event timeline (illustration only; times are invented)**

| Seq. | Time (IST) | Event | Evidence source | Notes |
|---|---|---|---|---|
| 1 | Day 1, 09:12 | *Internship_Report_Final.docx* created in user E's Documents folder | File-system creation time | Creation time may reflect a copy |
| 2 | Day 1, 09:40 | File modified and saved | Modification time; document metadata | Content version V1 |
| 3 | Day 1, 10:05 | File copied to a USB device | USB connection record; shortcut/recent-file entry | USB serial recorded |
| 4 | Day 2, 18:30 | Restore point created (Volume Shadow Copy #2) | Snapshot creation time | Snapshot holds V1 |
| 5 | Day 3, 22:10 | File deleted and moved to Recycle Bin | Recycle Bin index entry | Deletion time recorded |
| 6 | Day 3, 22:12 | Recycle Bin emptied | Missing items, file-system journal | Original no longer visible |
| 7 | Day 3, 22:15 | Browser history cleared | Reduced or empty current history database | Older history may remain in snapshots |
| 8 | Day 4, 11:00 | Forensic image acquired and verified | Acquisition log | Evidence preserved |
| 9 | Day 5, 14:20 | Snapshot #2 opened; file recovered (V1) | Recovery workflow, Section 5 | Recovered version reflects state at Day 2, 18:30 |

**Interpretation:** the recovered file shows the document *existed in that state* at the snapshot time. It does not say who deleted it or why. If edits happened after the snapshot (between Day 2, 18:30 and Day 3, 22:10), the recovered version may differ from the final version.

# 7. Validating a recovered artifact before reporting

### Table 4. Validation checklist

| Check | Why |
|---|---|
| Image hash verified before recovery | Evidence integrity |
| Snapshot ID and creation time recorded | Source and time anchor of the artifact |
| Recovered item hashed; hash recorded | Later verification |
| Extraction repeated (or done with a second tool) with the same result | Rules out tool error |
| File opens correctly; header/signature matches the extension | Confirms it is a valid file, not corrupt data |
| Metadata compared (size, timestamps, author fields) with the live volume or other snapshots | Shows how the version differs |
| Corroborated with independent artifacts (Recycle Bin, shortcut/recent-file entries, journal, event logs) | One source alone is weak |
| Time zone and clock offsets checked | Avoids timeline errors |
| Alternative explanations considered (system clean-up, auto-deletion, restore, sync) | Prevents over-claiming |
| Tool names and versions documented | Reproducibility |
| Findings reviewed by a second examiner (peer review) | Quality control |
| Wording of the finding states only what the evidence supports | Defensible report |

# 8. Sources of error and alternative explanations

### Table 5. Possible errors and alternative explanations

| Area | Issue | Effect | Mitigation |
|---|---|---|---|
| Timestamps | Wrong system clock or time zone | Events appear shifted or out of order | Record clock offsets; normalise to one time zone |
| Timestamps | Copy or move operations alter times | Creation time may not mean original creation | Compare with other artifacts and journals |
| Timestamps | Deliberate manipulation | False timeline | Cross-check with logs and snapshots |
| Snapshot timing | Snapshot reflects a moment, not the final state | Recovered version may differ from what existed at deletion | State the snapshot time clearly |
| Snapshot timing | Snapshots deleted, expired or limited by storage | Missing history may be innocent or deliberate | Examine settings, event logs and storage limits |
| Deletion | File removed by the system (clean-up tools, antivirus, updates), not the user | Wrong attribution of deletion | Check logs and tools installed |
| Deletion | Emptying the Recycle Bin or deletion by another account | Wrong person blamed | Correlate with logon events and profile |
| Browser artifacts | Cleared by routine settings (e.g., auto-clear on exit) | History absence is not proof of concealment | Check browser settings |
| Recovered data | Carved or partial files, overwritten blocks | Incomplete or incorrect content | Validate; report as partial |
| Tools | Parsing bugs or version differences | Wrong interpretation | Second tool, documented version |

| Area | Issue | Effect | Mitigation |
|---|---|---|---|
| Attribution | Shared account or remote access | Account is not a person | Seek corroborating evidence |

## Conclusion

Volume Shadow Copies can preserve earlier states of files and browser artifacts that were deleted or changed. A defensible finding needs careful selection and classification of evidence, a read-only recovery from identified snapshots, event-based reconstruction of the sequence, and validation by independent checks. Findings should describe what existed at a known snapshot time and clearly state timestamp and attribution limitations.

### References


1. Casey, E. (2011). *Digital Evidence and Computer Crime*, 3rd ed., Academic Press.
2. Carrier, B. (2005). *File System Forensic Analysis*. Addison-Wesley.
3. Kent, K. et al. (2006). *Guide to Integrating Forensic Techniques into Incident Response*, NIST SP 800-86.
4. Microsoft. *Volume Shadow Copy Service* (Microsoft Learn documentation).
5. libyal project. *libvshadow: Volume Shadow Snapshot (VSS) format documentation*.
6. ISO/IEC 27037:2012. *Guidelines for identification, collection, acquisition and preservation of digital evidence*.
7. ILookIX documentation (Perlustro LP): confirm the exact Event, Cloud and Lead Analysis features in your installed version.

# 2.5 MULTI-SOURCE INVESTIGATION OF A SUSPECTED CYBER INCIDENT

**Selected scenario:** Unauthorized access to the college online assignment-submission portal using a student account, with evidence spread across a lab PC, browser, email, messaging, an Android phone and cloud services.

**Assumptions**

A1. The student account is referred to as **account S**. The incident is an unauthorized login to the portal followed by export of a file of student marks.

A2. Written authorization exists for the lab PC, the phone and requests to the portal administrators. Access to third-party cloud, email and messaging providers needs separate lawful process.

A3. The PC runs Windows 10 with a Chromium-based browser; the phone is an Android device; email is webmail; messaging is a mobile chat app. All are generic.

A4. All times are IST and all artifacts, names and IP addresses are **hypothetical**. No real findings are claimed.

## 1. Potential evidence from Windows and other operating systems

**Table 1. Operating-system evidence sources**

| System | Evidence | What it can show |
|---|---|---|
| Windows | Security/system event logs | Logon and logoff of accounts, remote sessions, service and policy changes |
| Windows | Registry hives (NTUSER.DAT, SYSTEM, SOFTWARE) | Programs run, recent files, USB device history, installed software, time zone, Wi-Fi networks |
| Windows | Prefetch, shortcut (LNK) files, Jump Lists | Program execution and files opened by each user |
| Windows | Recycle Bin, file-system metadata, journal | Deleted files, creation/modification times |
| Windows | Page file, hibernation file, RAM capture | Remnants of passwords, URLs, chat text, running processes |
| Windows | Scheduled tasks, startup entries, installed applications | Persistence or tools used |
| Linux (e.g., portal server or a student's second device) | Authentication and web-server logs, shell history, cron jobs, home directory | Logins, commands run, files accessed |
| macOS (if a student device) | Unified logs, property lists, file-system event records, keychain | Application use, logins, file activity |
| Server OS / web application | Web-server access logs, application logs, session records | Source IP, user agent, URL and time of each request made as account S |
| Mobile OS (covered in Section 4) | App databases, system logs, Wi-Fi records | User activity and location of the phone |

## 2. Browser artifacts and why they must be interpreted carefully

### Table 2. Browser artifacts

| Artifact | What it may reveal |
|---|---|
| History (URLs, titles, visit times, visit counts) | Sites and pages visited, including the portal login page |
| Downloads list | Files downloaded, source URL, save location, time |
| Cache and temporary files | Copies of pages, images and sometimes file content |
| Cookies and session data | Logged-in sessions, site preferences, account identifiers |
| Form data and saved logins | User names typed, saved credentials (stored encrypted) |
| Search terms | Queries entered into search engines |
| Bookmarks, extensions, session-restore files | Interests, tools in use, tabs open at last close |
| Synced data | History and passwords synchronised from other devices |

### Why interpretation needs care

- A visit record shows a page was loaded, not that the user read it or intended it (redirects, pop-ups, background tabs, link pre-loading).
- A shared computer or shared browser profile makes it hard to say who was browsing.
- Synchronisation can make history from another device appear on this one.
- Browsers store times in their own formats (e.g., microseconds since 1 January 1601 in Chromium-based browsers). Wrong conversion shifts the timeline.
- Private browsing, history clearing and automatic clean-up reduce evidence; absence of history is not proof of hiding.
- Cached or saved data may predate the incident.

## 3. Email and messaging evidence and correlation

### Table 3. Email and messaging evidence and its correlation value

| Evidence type | Details | How to correlate with other artifacts |
|---|---|---|
| Email headers | Sender, recipient, date, message ID, relay (Received) lines, sometimes the originating IP | Compare originating IP and time with portal logs and campus DHCP/Wi-Fi logs |
| Email body and attachments | Content, file names, embedded links | Hash an attachment and compare with the file found on the PC or in the browser downloads |
| Mailbox and cache files (e.g., local mail stores, webmail cache) | Sent, deleted and draft items | Match sent times with browser sessions and logon events |
| Calendar and contacts | Meetings, contacts, relationships | Show associations and context |
| Chat database on phone/PC | Messages, timestamps, sender/receiver IDs, group names | Compare message times and mentions of the file or login with the timeline |
| Shared media and attachments | Images, documents sent through chat | Hash and compare with PC and cloud files |
| Call logs and voice/video metadata | Who communicated with whom and when | Show coordination between persons |
| Provider-side records | Login history, IP addresses, message metadata (through lawful process) | Confirm device and location of access |

Headers can be forged, and sender names are easily spoofed, so headers should be validated against server-side records where available.

## 4. Forensic considerations for the mobile phone

### Table 4. Mobile examination considerations

| Consideration | Action / reason |
|---|---|
| Legal authority and scope | Phones hold large amounts of personal data, so examine only what the authorization covers |
| Isolation | Prevent remote wiping and new data: airplane mode if the device is on and unlocked, or a Faraday bag. Record the power and lock state |
| Do not power cycle unnecessarily | Powering off may lock the device and lose volatile data and keys |
| Acquisition level | Logical, file-system, physical or chip-level methods are possible, depending on device, operating system version and lock status. State which was used and why |
| Encryption and passcode | Modern phones use strong encryption. Without the passcode, extraction may be limited |
| Hashing and documentation | Record device identifiers (e.g., IMEI), tool and version, acquisition type and hash of the extraction |
| Cloud and backups | Phone data may sync to cloud accounts, which need separate authority |
| Time handling | Record the phone's clock and time zone against a reference time |

### Challenges with mobile evidence

- Wide variety of devices, operating system versions and applications, with frequent updates.
- Strong encryption and locked devices.
- Application data in protected locations, in changing formats; tool support may lag behind new app versions.
- Disappearing or deleted messages and remote-wipe risk.
- Data mixes personal and case-related content, so privacy handling is needed.
- Evidence may also live in backups and cloud storage, outside the device.

## 5. Internet and cloud-based evidence sources

### Table 5. Internet and cloud sources

| Source | Evidence | How obtained / limits |
|---|---|---|
| Portal web-server and application logs | Login attempts, source IP, user agent, accessed pages and file exports for account S | Institution's own administrators; preserve and hash copies early |
| Authentication/single sign-on and multi-factor logs | Successful and failed logins, device and location data | Institution or identity provider |
| Campus network logs (DHCP, Wi-Fi/RADIUS, firewall, proxy, DNS) | Maps IP address to device (MAC) and time; shows connections | Institution; retention periods vary |
| Internet service provider records | Subscriber details for an IP address | Lawful request through proper authority |
| Email provider records | Account login history, IP addresses, message metadata | Lawful process; provider retention varies |

| Source | Evidence | How obtained / limits |
|---|---|---|
| Cloud storage and productivity services | Upload and sharing logs, file versions, deleted items | Account owner's export or lawful process |
| Messaging and social media providers | Account records, message metadata | Lawful process; end-to-end encryption limits content |
| Third-party and archive sources | Public posts, cached pages, archived copies | Open-source collection; document method and time of capture |

Cloud evidence should be preserved early, because retention periods are limited and users can delete data. Data may sit in other jurisdictions, which complicates access.

### Table 6. Cross-source correlation matrix (artifacts, times in IST)

| ID | Source | Hypothetical artifact | Time | Links to | What the link shows | Caution |
|---|---|---|---|---|---|---|
| C1 | Internet/cloud (portal log) | Successful login as account S from IP *a.b.c.d*; later request to export a marks file | 21:03; 21:15 | C2, C3, C4 | The account was used from this IP | Account is not a person |
| C2 | Internet (campus DHCP/Wi-Fi log) | IP *a.b.c.d* leased to MAC of the lab PC | 20:58 | C1, C3 | Ties the IP to a specific computer | Check lease times and clock offset |
| C3 | Computer (Windows event log) | Logon by a local user account on the lab PC | 20:56 | C2, C4 | Someone was logged on locally at that time | Shared lab account possible |
| C4 | Browser (history and downloads) | Portal login page visited, then *marks_export.csv* downloaded | 21:03; 21:15 | C1, C5 | Browser on this PC fetched the file | Visit is not proof of who typed |
| C5 | Email (webmail sent item) | Message from S's mailbox with attachment of the same file name | 21:30 | C4, C6 | File left the PC by email | Header time vs. server time |
| C6 | Computer + email | Hash of the attachment equals the hash of the downloaded file | n/a | C4, C5 | Same file content in both places | A hash match proves content identity only |
| C7 | Mobile (chat database) | Message to a contact referring to "the file" | 21:35 | C5 | Possible sharing or coordination | Message wording is ambiguous |
| C8 | Mobile (Wi-Fi connection record) | Phone connected to the same campus Wi-Fi | 20:45 | C2, C7 | Phone and PC were on the same network | Presence on a network is not use |
| C9 | Cloud (storage activity log) | Upload of the same file from IP *a.b.c.d* | 21:40 | C1, C6 | Possible onward copy | Needs provider confirmation |

## 6. Major challenges in multi-device and cloud forensics

### Table 7. Challenge analysis

| # | Challenge | Why it matters | Mitigation |
|---|---|---|---|
| 1 | Data volume and variety | Many devices and formats make review slow and error-prone | Use triage, filtering and timelines; document criteria |
| 2 | Encryption and locked devices | Content may be unreadable | Seek lawful key/passcode access; rely on metadata and server-side records |
| 3 | Cloud data location and jurisdiction | Data may be held abroad or by third parties | Early preservation requests; lawful process; legal advice |
| 4 | Limited control and retention | Providers keep logs for limited periods; users can delete | Send preservation requests quickly |
| 5 | Time synchronisation | Each source has its own clock, time zone and format | Normalise to one reference; record offsets |
| 6 | Chain of custody for remote data | No physical item to seal; data may change | Record method, time, account and hash of exports |

| # | Challenge | Why it matters | Mitigation |
|---|---|---|---|
| 7 | Authenticity and integrity of provider-supplied data | Hard to verify what the provider returns | Document source and request method; cross-check with other sources |
| 8 | Shared accounts and multiple users | Account does not equal person | Corroborate with physical presence, devices and logs |
| 9 | Privacy and legal constraints | Devices and cloud accounts hold personal data | Limit scope to authorization; follow data-protection law |
| 10 | Anti-forensic actions | Deletion, wiping, VPN use, private browsing | Look for traces in backups, logs and other devices |
| 11 | Tool and version compatibility | Apps and formats change fast | Use validated tools; test with a second tool |

## 7. Final investigation conclusion (based on the hypothetical artifacts above)

This conclusion illustrates the structure of a final summary. Real conclusions depend on real evidence.

**Table 8. Final investigative summary**

| Category | Content |
|---|---|
| Direct evidence | Portal log (C1) records account S logging in and exporting the marks file. Webmail sent item (C5) records a message with that file attached from S's mailbox. Both show activity by the *account* directly |
| Corroborating evidence | DHCP/Wi-Fi log ties the IP to the lab PC (C2). Windows logon (C3) and browser history and download (C4) place the activity on that PC. Hash match (C6) shows the exported and emailed files are identical. Phone Wi-Fi record (C8) and chat message (C7) are consistent with the same period. Cloud upload (C9) suggests an onward copy |
| Unresolved questions | Which person was at the keyboard (shared lab account or shared credentials). How the credentials were obtained, and whether the account was compromised. Whether the phone user and the PC user are the same person. Whether deleted messages or files exist. Provider-side confirmation of the cloud upload. Clock offsets between sources |

**Conclusion :**

The combined artifacts support, with moderate-to-high confidence, that account S was used from the lab PC to access the portal, export a marks file and send it onward. They do not, by themselves, identify the individual operating the PC. Further steps: obtain provider records through lawful process, review lab access records, and interview relevant persons under proper procedure.

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